EXCLUSIVE FEATURE

SHAPING THE FUTURE

Sir Michael Marmot's Vision for Population Health Education

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PHOTO
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Being a population health student at UCL it is impossible not to be aware of Professor Sir Michael Marmot's body of work and the enormous effect he has had, and continues to have, on population health in the UK and worldwide.

We wanted to learn more about his time at UCL, and his advice for students now beginning their journeys in population health. Our Editor-in-Chief Tanya Singh sat down with Professor Marmot and did just this.

COULD YOU TELL US ABOUT YOUR TIME AT UCL?

Upon being appointed in February of 1985, Professor Marmot had two notable reactions to his appointment. One was from Sir James Lighthill, then Provost and President of UCL saying:" 'Your discipline is going to be as important to medicine in the coming year as Physiology was to meds at the beginning of the 20th century'".

And an opposing view from the medical school: "At the time it was called community medicine and it was, I think, fairly typical of medical schools everywhere: 'What we do is first, last and foremost with the patient and then the basic science underpinning what we do with the patients. What is this? Community meds and public health? What is this? Why? Why do we need that? In a medical school?'"

But despite this lack of belief in the role of population health from the medical school, Professor Marmot's research team began to grow in number - and in grants: "I started to bring in research grants from the Research Council, British Heart Foundation, from the US National Institutes of Health". And then the attitude of the medical school began to change: "this isn't this fluffy, airy thing you know" and a realisation that "It doesn't take away from the importance of cognitive neuroscience, molecular biology. But it's another branch of science that has real implications in the wider world".

With the faculty being established, by 2011 the team had grown from 7 to 170 people. Professor Marmot had truly made a change in how public health was perceived within UCL, from being 'community medicine' and just an accessory to the Medical School, the Faculty of Population Health Sciences is now a world leading body in analysing population health and improving population health.

WHAT IS THE ROLE OF STUDENTS IN GLOBAL HEALTH?

"Students' main function, of course, is to get educated but, education doesn't just mean what you get in lectures or in the syllabus: getting educated can mean hands on experience" and Professor Marmot uses as an example of "terrific enthusiasts, students who were concerning themselves with global health, firstly to get educated, but secondly to take action; whether it was organising policy dialogues or thinking about how they could get involved in activities in low and middle income countries particularly."

WHAT ARE THE SKILLS STUDENTS SHOULD DEVELOP TO BE FUTURE LEADERS IN POPULATION HEALTH?

"This is my own view (for what it's worth!) that you want to be involved in activism one way or another". Given how much work Professor Marmot has done to improve population health, we trust him! We know there is still room for more action to push governments and communities to continue to work to improve public health.

And that means: "You need to be at the very least informed consumers of evidence... and have general research skills, particularly in population health". And in the Faculty of Population Health Sciences, we do just that: robust critical thinking is a skill honed by students across the faculty; the strengths and weakness of any research discussed are evaluated so students become such "informed consumers" of research.

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The sphere of healthcare is one that can be influenced by political ideologies. One should, argues Sir Michael, be able to analyse the evidence and reach a balanced conclusion rather than aligning oneself based purely on ideological thinking.

Furthermore, Professor Marmot believes it is vital to engage with issues such as climate change and health disparities. Professor Marmot builds a strong case for activism and political understanding along with a critical, evidence-based perspective to create positive, sustainable and global changes to health outcomes.

Professor Marmot also sees an interdisciplinary view as key: "it's not just biomedical knowledge that you need or technical issues like 'the vaccines work' but you need to start to understand issues of macroeconomics, economic arrangements, and how they influence health". Developing well rounded population health scientists has been a priority of the faculty, with students on the BSc Population Health Science course being encouraged to take modules like health economics, to the emergence of entirely new institutes like the Global Business School for Health.

However, this focus on collaboration does not limit itself to the faculty level: students have developed their own skills to adapt to working in a range of industries. As you will see in the rest of the magazine, there are many interests being explored by staff and students in the Faculty of Population Health.

WHAT DO YOU HOPE TO SEE FROM THE POPULATION HEALTH SCIENCES MAGAZINE?

Professor Marmot stated his desire to see the magazine "flourish" as a forum to facilitate the exploration of different health topics, including health inequality, climate change and communicable diseases. The articles have been written in a way to foster discussion and encourage the sharing of ideas.

The challenge set by Sir Michael was to provide something distinctive to what other established journals such as the British Medical Journal or The Lancet, offer. This student-led magazine aims to achieve just this by bridging the gap between academia and hands-on involvement, inspire critical thinking, facilitate discussion and instigate action among future healthcare professionals.

GENOMICS

TO TACKLE TUBERCULOSIS

A spotlight on UCL initiatives in Peru

Led by Dr Louis Grandjean, the UCL team comprises two UK clinical researchers, one UK PhD student, and numerous collaborating researchers in Peru. Based at Universidad Peruana Cayetano Heredia (UPCH), a leading biomedical teaching and Research Institute located in Lima, Peru, our goal is to utilise genomic analyses of Mycobacterium tuberculosis (Mtb) to explore associations of Mtb with clinical pathology for the development of novel treatment approaches.

Tuberculosis and Peru

Tuberculosis (TB) is the leading cause of death from infectious diseases worldwide, with an estimated 1.3 million deaths and an estimated global incidence of 10.6 million new (reported) TB cases in 2022. Multidrug-resistant tuberculosis (MDR-TB) represents a substantial portion of tuberculosis cases and has rapidly developed into a significant global public health concern. (1) MDR-TB is characterised by its resistance to isoniazid and rifampicin, both fundamental first-line drugs used in tuberculosis treatment. The second-line therapies required to treat MDR-TB tend to cause more disabling side-effects, thereby leading to poorer treatment outcomes. (2)(3)

Peru is one of thirty countries identified by the World Health Organization (WHO) as having a significant/major TB burden. The problem is further complicated by the high prevalence of MDR-TB; the new infection rate of TB in Peru is 151 per 100,000 people each year with an MDR-TB incidence rate of 8.3 per 100,000. (1) The WHO has set a goal to eliminate the TB epidemic by 2035. In pursuit of this objective, there is a collective need for innovation and the development of novel strategies to address the challenges posed by TB. Our work in Peru is spearheading research in the field of TB by applying innovative and advanced genomic strategies.

Photo: The UCL and UPCH team by the entrance of the laboratories at UPCH

Genomics

In recent years, whole-genome sequencing has been a pivotal tool in the study and control of tuberculosis. Advanced next-generation sequencing (NGS) technologies enable high-throughput whole genome sequencing of Mtb, which provides invaluable insights into its genetic diversity. (4) Such information is crucial for understanding the disease's evolution and transmission as well as identifying drug-resistant strains, thereby facilitating the development of targeted treatment strategies.

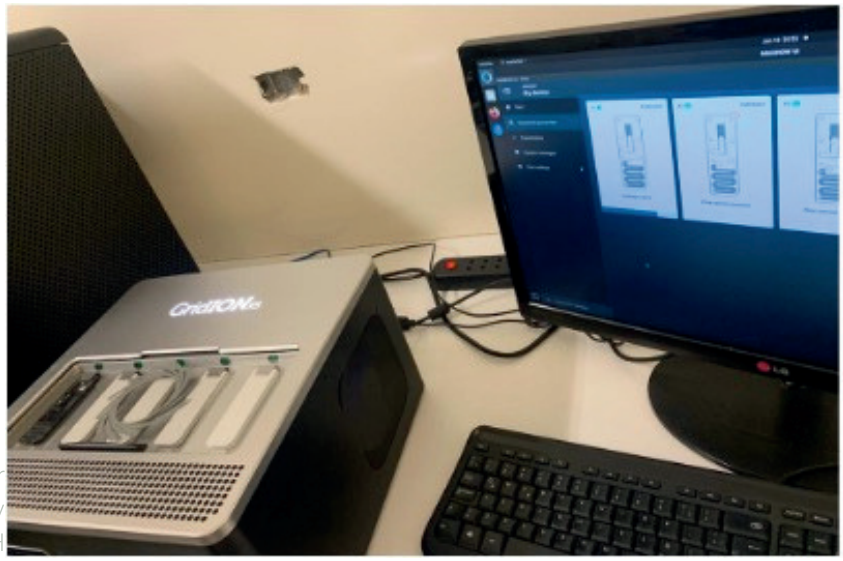

Photo: Nanopore GridION sequencer located in the Laboratory of Bioinformatics at UPCH

Mycobacterium Tuberculosis DNA Extraction for Nanopore Sequencing

The development of targeted drugs, more effective vaccines, and improved diagnostic tools is underpinned by analysing the genetics of the bacteria to identify specific genetic mutations that confer resistance to antibiotics or influence virulence. To perform genetic analysis of Mtb, genomic DNA must first be extracted from within the cultured bacterial cells, which is exceptionally complex due to its thick lipid and mycolic acid cell walls. Traditional methods of DNA extraction rely on harsh chemicals or abrupt mechanical beating to liberate the DNA. This results in short DNA strands and chemical contaminants in the final DNA. This is problematic for downstream Nanopore sequencing analysis which excels at sequencing long strands of DNA and requires a high purity of DNA input. Therefore, our aim was to optimise an extraction method for Nanopore sequencing to release sufficient DNA, whilst preserving the DNA's integrity and purity.

Current projects

1. Mycobacterium Tuberculosis DNA Extraction for Nanopore Sequencing
2. The Effect of Mycobacterium Tuberculosis Genotype on Clinical Outcome
3. Association between Mycobacterium Tuberculosis Genome and Radiological Pathology on Chest X-Ray (CXR)

Despite the challenges of Mtb, we successfully developed a refined technique to extract high-purity, lengthy DNA strands suitable for Nanopore sequencing. In this process, we also identified limitations to conventional DNA extraction techniques, particularly in their compatibility with Nanopore sequencing. This breakthrough will serve as the cornerstone for future experiments employing Nanopore sequencing, essential for vaccine and diagnostics research.

Photo: A member of the team culturing Mtb in the P3 laboratory at UPCH

The Effect of Mycobacterium Tuberculosis Genotype on Clinical Outcome

Variations in the genetic code of Mtb can affect how infectious the disease is, which antibiotics it is resistant to, and the extent of disease it causes in the patient. These aspects are important to consider when predicting the clinical outcome of patients with TB.

Mtb bacterium has been co-evolving with humans for tens of thousands of years. Its genome today is highly varied and encompasses multiple lineages, strains, and mutations. The interaction between Mtb and its human host is influenced by the pathogen genome. As a result, it is hypothesised that differences in the Mtb genome may help to predict patients' outcome from treatment, namely, whether their infection would be resistant to antibiotics, resulting in a relapse. (5) Currently, we are conducting a retrospective population study to investigate the correlation between Mtb genotypes and patient outcomes, focusing on the conversion from drug-susceptible to multidrug-resistant (MDR) infections. With better knowledge about which types of infection will gain resistance over time, doctors can adapt the treatment scheme of the patients to anticipate and avoid future drug resistance, ensuring better outcomes for the patients, and helping reduce the general level of antibiotic resistance in the population.

Association Between Mycobacterium Tuberculosis Genome and Radiological Pathology on Chest X-Ray (CXR)

CXR, a radiological evaluation of the lungs, is the primary diagnostic method for pulmonary TB and is widely used in high and middle-income settings. Identifying bacterial genomic markers linked to radiological severity and bacterial load could facilitate contact tracing of high-risk individuals for secondary disease transmission. This insight could also help personalise treatment regimens based on the pathogen's genome.

In 2018, we conducted the first study to compare the Mtb genome to radiological pathology on CXR. We performed whole genome sequencing on Mtb isolates from 103 patients with drug-resistant TB in Lima between 2010 and 2013. Our research interestingly identified two mutations significantly associated with more widespread radiological pathology and one mutation with reduced pathology. Our findings were published in the BMJ Thorax journal for dissemination within the scientific community. Prospective studies are also warranted to determine whether mutations with increased pathology also predict the spread of drug-resistant TB; thus,we are currently undertaking a prospective cohort study using newly confirmed drug-susceptible pulmonary tuberculosis cases.

Summary

The UCL team, together with our esteemed peers in Peru, are pioneers in TB research on many fronts. By delving into the genome of Mtb to understand transmission patterns and clinical significance, we strive to drive advancements in the diagnosis, treatment, and control of tuberculosis.

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COST-BENEFIT ANALYSIS AIMS TO MIMIC THE PRIVATE SECTOR but does it?

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In today's societies, cost-effectiveness assessments (CEA) and its main tool, Cost-Benefit Analysis (CBA), are critical to make decisions. This short piece seeks to argue that we can do better. Updating CBA should be a major topic of research in economics.

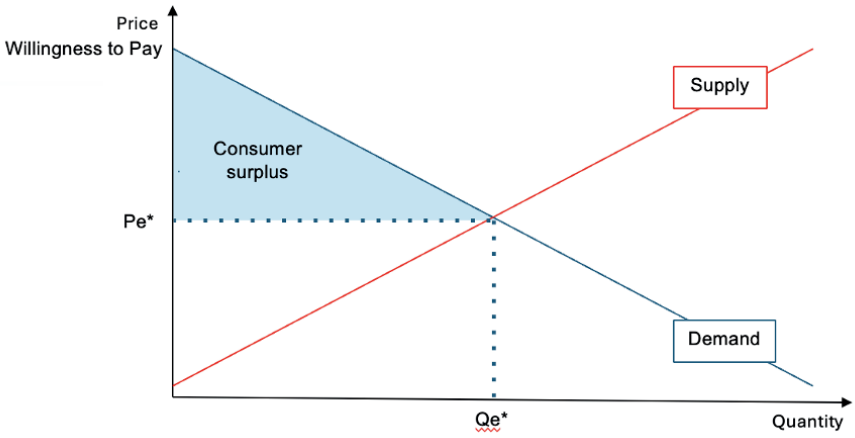
The intuition behind Cost-Benefit Analysis was first introduced in 1844 by Jules Dupuit (Dupuit, 1952, 1844). The technique was then formalised by Marshall in 1893 (Marshall, 1893). In 1972, a major book in the field set up the practice (Dasgupta & Pearce, 1972). Since then, many refinements have been made (e.g. Nesticò et al., 2023). Below is a brief introduction to the technique.

INTRODUCTION TO COST-BENEFIT ANALYSIS TECHNIQUE

CBA is aiming at mimicking the private sector market ("the market"), because, according to economic theory, the market is Pareto optimal. In lay terms, Pareto optimality means that no other allocation of resources is possible to improve at least one individual's situation without harming at least one individual. Therefore, the public sector aims to get to a Pareto optimal allocation of resources by mimicking the private sector framework

In the private sector, theoretically, supply and demand meet to obtain the equilibrium defined by an equilibrium price and quantity.

DIAGRAM 1



In Diagram 1, a simple representation of a market for a good called B shows the price and quantity at the equilibrium and . The solid blue triangle is what Marshall conceptualised as the consumer surplus. The consumer surplus is what all those consumers who are willing to pay more than the equilibrium price for B but only have to pay the equilibrium price: , gain[1] .

In CBA terms, the consumer surplus is called the Net-Benefit:

$$WTP_B - P_{eB} = Net\ Benefit = Consumer\ surplus$$

Where B is the good of that market.

OPPORTUNITY COST

Economists then introduce another concept: opportunity cost. The opportunity cost is the Net-Benefit of the alternative that is not chosen. Say that we compare goods B and A. In CBA, we subtract the Net-Benefit of the alternative A from the Net-Benefit of good B, which results in the Incremental Net-Benefit:

$$INB: WTP_B - Cost_B - (WTP_A - Cost_A)$$

This equation reflects the theoretical decision that consumers make in the private sector. The willingness-to-pay (WTP) is the monetary translation of the characteristics of a product. Indeed, the idea is quite simple and makes sense: When buying new electronic equipment (e.g. a washing machine) one is going to balance the characteristics of each washing machine against their cost/price, which gives the Net-Benefit of that machine. Suppose the Net-Benefit of washing machine B is much higher than the alternative's Net-Benefit. In that case, an informed, rational consumer will choose good B.

Basically, what CBA does, is apply simple economic reasoning to public investments. The consumer, like the public sector, shares the same constraint: the budget.

However, we argue that this reasoning is missing a key point in the funding of the private sector: **when firms invest, the budget may not be that constrained.**

Schumpeter (2017, 1934) has identified that new investments need funding; for the private sector, a bank lends money. McLeay, Radia & Ryland (2014) explain where most of the source of finance in the economy comes from: commercial banks. Commercial banks create new money, out of nothing, to fund businesses and individuals likely to stay clients (2014).

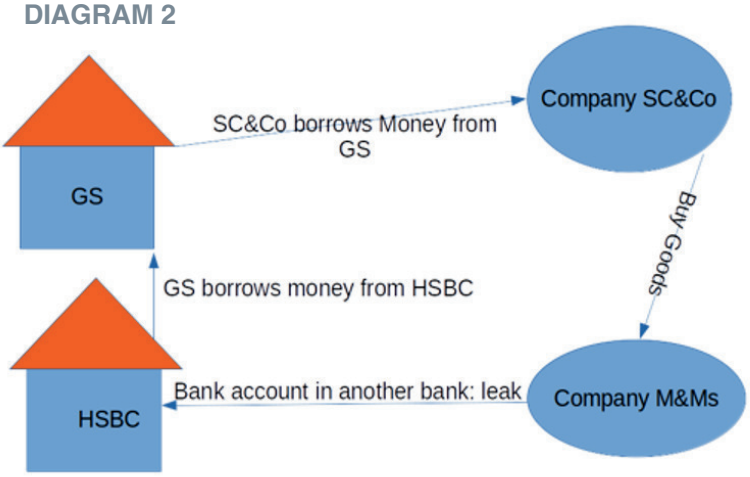
LEAK IN BANKING

Commercial banks can 'virtually' create as much money as they wish. However, they are constrained by leaks to other banks. Say that we have two banks and two companies[1]...

On a typical day, a company SC&Co borrows money from a bank: GS. With that money, SC&Co buy goods from a company M&Ms. That latter company has a bank account with HSBC. At the end of the day GS is short of money and its books are unbalanced because the money it lent to SC&Co ended up in the accounts of HSBC; GS will have to borrow money and HSBC, that day has a surplus so it can lend the money to GS. This is what is called a leak in banking. Because of this, banks lend (create) money in as much as the rest of the other banks. Banks mainly compete to get more clients, so that they have less leaks. GS would not have a leak in the following two cases:

1. If the same day HSBC lends to a company that buys from SC&Co.
2. If M&Ms was a client of GS.

DIAGRAM 2



Commercial banks do not compete for savings, they compete for clients, specifically successful clients, because they are likely to stay as customers.

The Central Bank (CB) plays a minor role in the quantity of money created by commercial banks. It sets an interest rate at which commercial banks can refund themselves. Instead of borrowing money from HSBC, GS could borrow it from the CB at the interest rate it has set up.

PRIVATE AND PUBLIC SECTOR DIFFERENCES

The public sector does not have access to this money. After World War II, most Western states rebuilt their economy with money from the Central Bank. However, during the 1920s and after WWII, the rationale for a CB independent from governments went its way to forbid CB lending directly to governments (Crowe & Meade, 2007), as it was assumed/advanced that with an independent CB, there would be less inflation. It led to a tremendous reduction in inflation and made it almost impossible to get money from the CB making the economy much more stable. It was argued then that States should have a hard budget constraint: it was (and still is) a great stabilizer.

The goal here is not to say that states should get funded by the CB without any constraints. However, not lending to States, is making the funding of public services reliant only on tax. The budget of the State became constrained and so did public services. Let us take a simple analogy here: money creation is like learning to ride a bike. States and the CB start riding without stabilizers, they fall into inflation; by putting stabilizers, inflation would be low for quite some time.

We argue that we need to change the stabilizer: the budget constraint should be replaced. The constraint with the stabiliser is that the budget is fixed, namely, it depends on the quantity of money collected through tax. However, the private sector has access to more money being created without the same constraint.

[1] The example is our understanding of banking from reading different articles (McLeay, Radia & Ryland, 2014; Werner, 2014).

ALTERNATIVE SOLUTIONS

Combes & Becker are proposing that we change the constraint to an inflation-based constraint (Combes & Becker, 2022). Instead of the CBA, economists need to consider an Inflation-Benefit Analysis or IBA.

The public sector would not compete for the money because its aim is not to sell a product and therefore outpace competitors; the aim is to provide public services. However, the private sector does compete for funds and banks do face constraints, namely, that of leaks. The money they create may end up in others' hands and therefore they can only create money in the same percentages as competitors; they may take risks by creating more money than competitors to gain clients.

The public sector needs another constraint to fund its public services. What money creation does is that it anticipates the creation of value. Public services are creating value that needs to be anticipated, such as in education. In healthcare, prevention and public health are too often not funded or inadequately funded. In Costa-Rica, public health teams wander around the country to meet patients even when they are not sick[1] . Costa-Rica has one of the highest life expectancies (as high as the US) at much lower expenditure. Funding for prevention and public health needs to anticipate the future rise in health expectancy and money should be made available for that, anticipating future outcomes and consequences of healthcare.

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